



CL 299 presentation 1

LNP Manufacturing and Process Simulation

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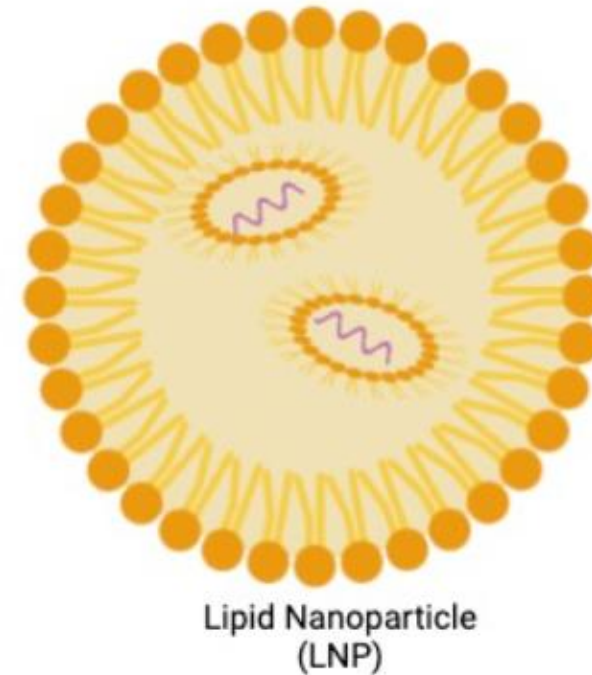
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MOTIVATION AND BACKGROUND

- Rapid success of mRNA vaccines. (Eg: Pfizer-BioNTech and Moderna).
- Directly injected mRNA would be rapidly degraded by enzymes in the body before it can reach and enter cells.
- LNPs are the industry-standard delivery system for mRNA.
- Major challenges: scalability, reproducibility and controllable manufacturing of LNPs.
- Commercial manufacturing requires tight control on size and uniformity.

Lipid Nanoparticles (LNPs) ?

- Tiny, spherical particles made of lipids.
- Small size - (10nm - 100nm).
- Widely used in vaccines.
- They encapsulate and protect fragile molecules like mRNA to reach and enter in cells efficiently.
- Human cell membranes are also made up of lipids so body is already familiar with LNPs.



Before LNPs → Liposomes

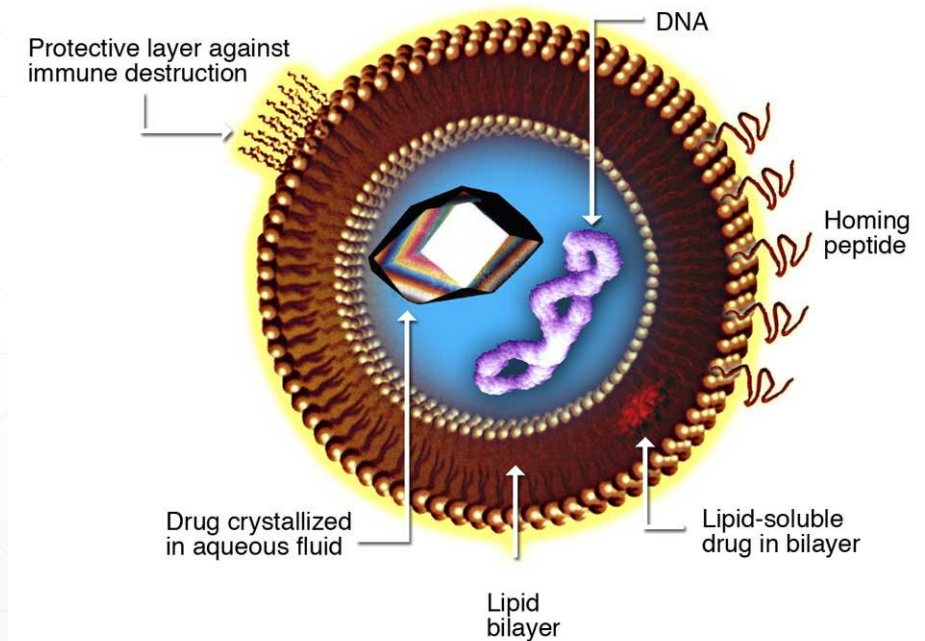
Liposomes are spherical vesicles composed of one or more phospholipid bilayers enclosing an aqueous core.

- Liposomes were the first clinically successful lipid-based nanocarriers.
- Widely used for small molecules, proteins, and early nucleic acid delivery.

Why liposomes were originally used for nucleic acids :-

- Phospholipid bilayers are biocompatible
- Can encapsulate hydrophilic cargo (DNA/RNA) in aqueous core
- Protect cargo from enzymatic degradation

Liposome for Drug Delivery



Disadvantages of liposomes for mRNA delivery

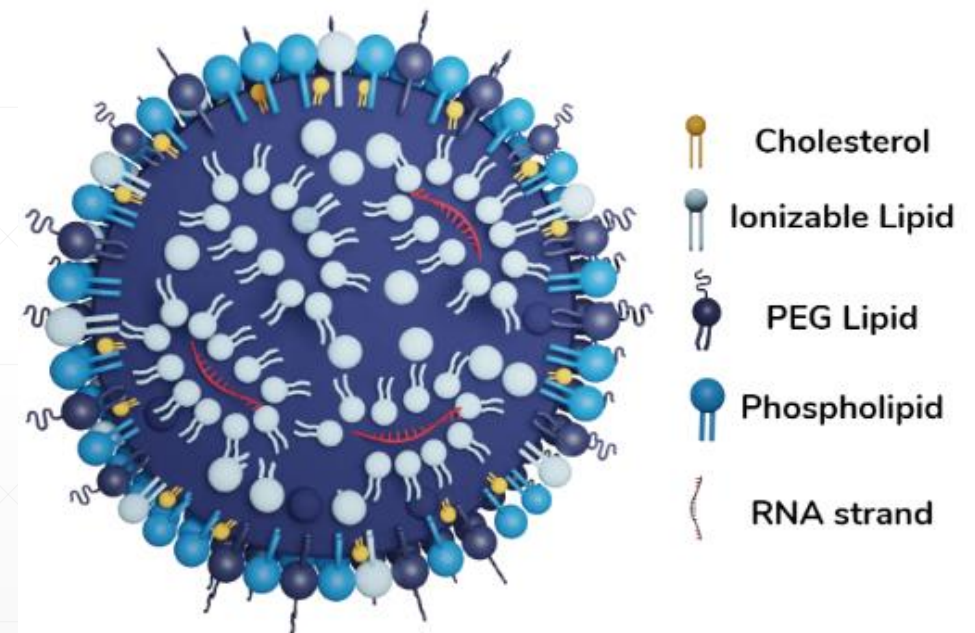
- Low mRNA encapsulation efficiency because liposomes rely on passive loading and lack strong electrostatic interaction with nucleic acids.
- Poor endosomal escape, resulting in degradation of mRNA before it reaches the cell.
- Limited scalability.

Why the field shifted from Liposomes to LNPs?

- Very high mRNA encapsulation efficiency (>90%) due to electrostatic complexation between ionizable lipids and negatively charged mRNA.
- Efficient endosomal escape enabled by ionizable lipids that become protonated at low pH and disrupt the endosomal membrane.
- Excellent scalability and reproducibility through continuous manufacturing methods (ethanol injection, microfluidics, jet/impingement mixing) with tight control of CQAs.

COMPOSITION OF LNPs

- Ionizable lipid: binds RNA and enables endosomal escape.
- Phospholipid: provides structural stability.
- Cholesterol: improve membrane rigidity and integrity.
- PEG-lipid: prevents aggregation and improves circulation time.

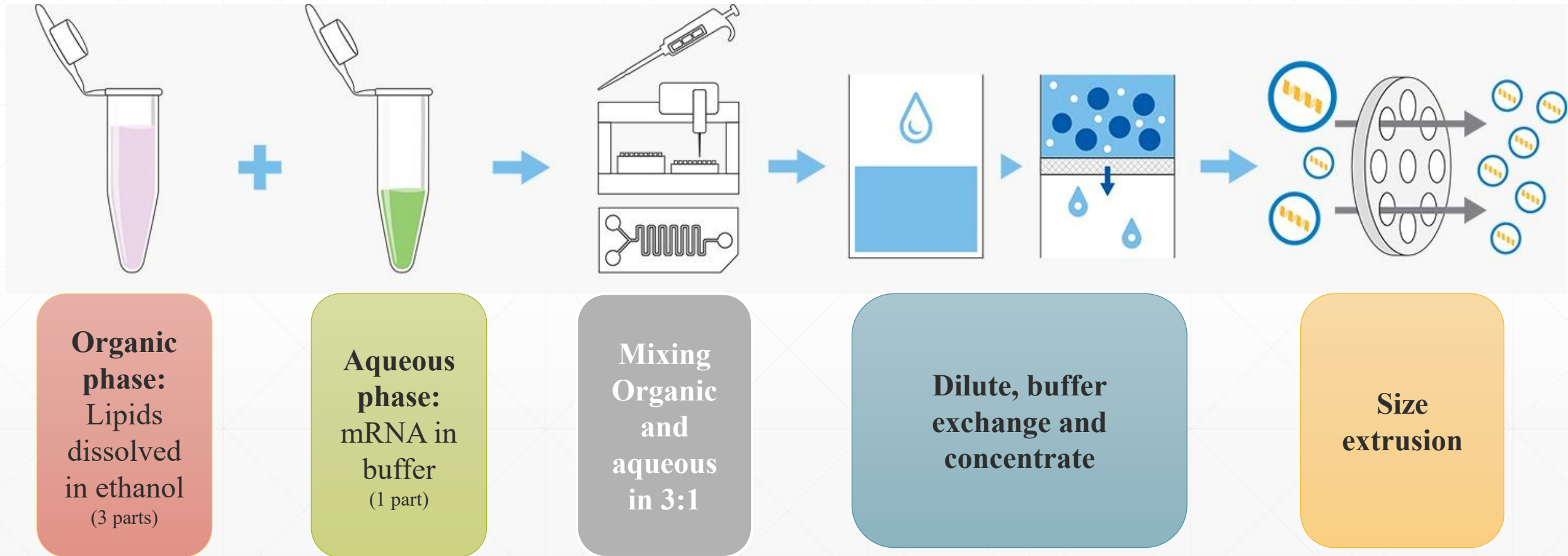


mRNA LNP (lipid nanoparticle) composition

Role of Ionizable lipids

- Ionizable lipids are neutral at physiological pH of blood → Low toxicity
- They are positively charged at acidic pH inside cells → RNA release inside cells.

LNP Formulation



LNP FORMULATION

- When Aqueous and organic phases are mixed, ethanol concentration suddenly drops.
- Rapid dilution of ethanol causes lipid oversaturation → lipids become insoluble.
- Lipids spontaneously self-assembled around mRNA.
- Process occurs within millisecond.

Hence, LNPs are formed.

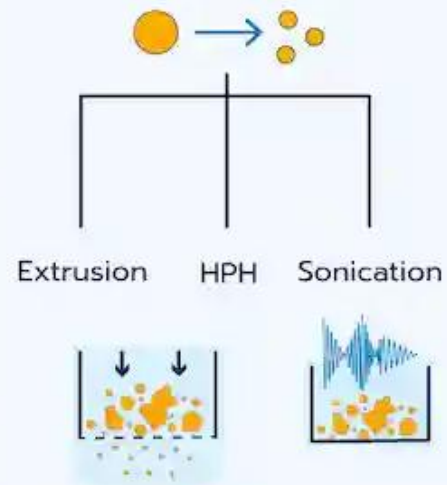
IMPORTANCE OF MIXING

- Particle size control and uniformity of LNPs are determined by mixing time.
- Mainly ensures a homogeneous temperature and mRNA concentration during formulation.
- Ensure high mRNA encapsulation efficiency → improve drug loading.
- The rate and manner of mixing directly affect self-assembly steps.
- Slow or uncontrolled mixing results in broad size distribution (high PDI).
- The faster the mixing the smaller the synthesized LNPs.

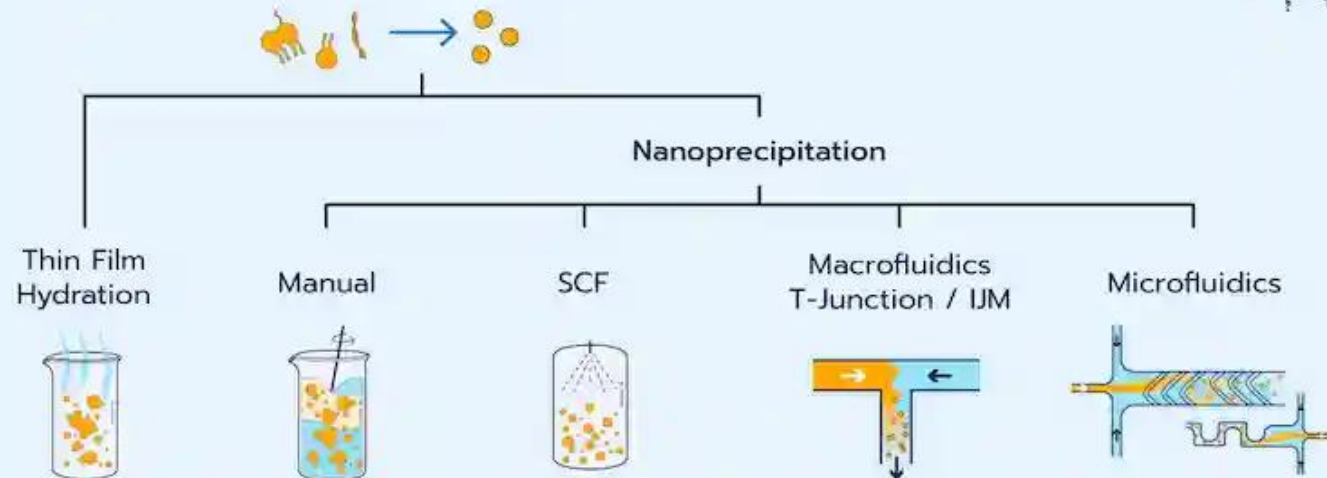
Methods to produce Nanoparticles & LNPs



Top-Down



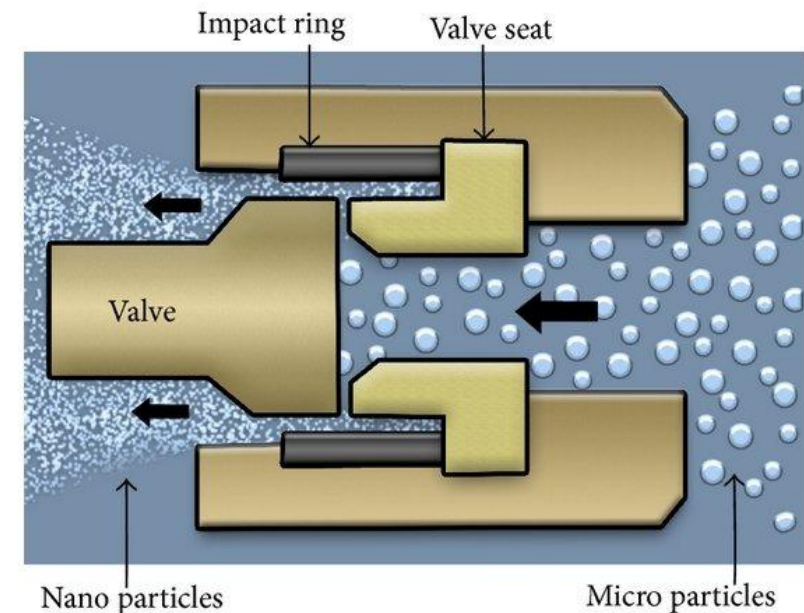
Bottom-Up



- **Top-down methods:** Also referred to as high-energy methods, these involve applying energy to break larger particles into smaller ones.
- **Bottom-up methods:** These rely on the self-assembly of lipid components to form homogeneous nanoparticle systems.

High-Pressure Homogenization (HPH)

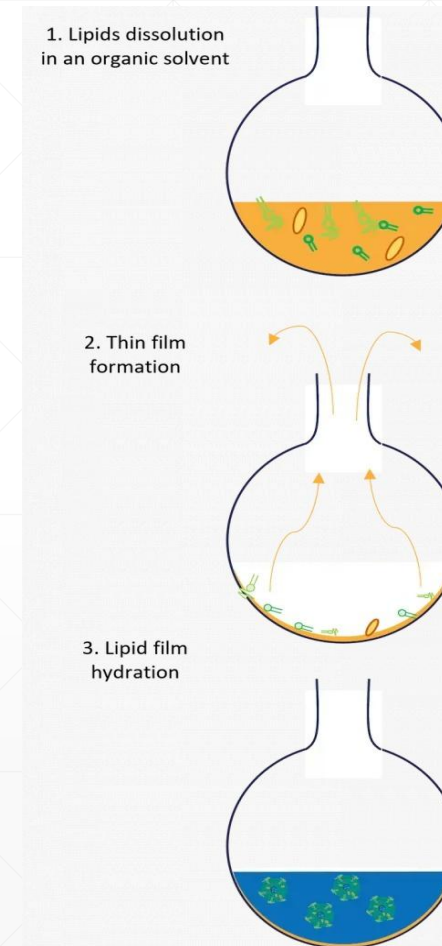
- Liquid is forced through a narrow valve at very high pressure, creating intense shear and turbulence for mixing.
- Droplet/particle size is reduced to the nanoscale due to shear, cavitation, and impact forces.
- Suitable for large-scale, continuous processing, but involves high energy input.



Bottom-up approach

Thin film Hydration

- Lipids are first dissolved in an organic solvent and dried to form a thin film on the container wall.
- The film is hydrated with an aqueous phase, causing lipids to self-assemble into vesicles/nanoparticles.
- It is a simple, widely used laboratory method, but offers limited control over particle size.



Manual method

- Lipid and aqueous phases are mixed manually using pipetting or simple stirring.
- The process offers minimal control over mixing intensity and uniformity.
- Results in poor reproducibility and broad particle size distribution.

Supercritical Fluid Methods (SCFs)

- Uses supercritical CO₂ as an antisolvent to mix with lipid solutions and induce nanoparticle formation.
- Enables rapid mixing with low residual organic solvent in the final product.
- Requires high-pressure, complex equipment, limiting routine use.

Macrofluidics

- Mixing occurs in millimeter-scale channels with controlled flow rates.
- Offers better control than batch/manual mixing but less precision than microfluidics.
- Suitable for higher throughput and easier scale-up compared to microfluidic systems.

Eg: Impingement Jet Mixing (IJM).

Microfluidics

- Fluids are mixed in micron-scale channels under precisely controlled flow conditions.
- Provides highly uniform and reproducible nanoparticles with narrow size distribution.
- Ideal for formulation screening and early-stage development, though scale-up can be challenging.

Eg: NxGen Mixing Technology.

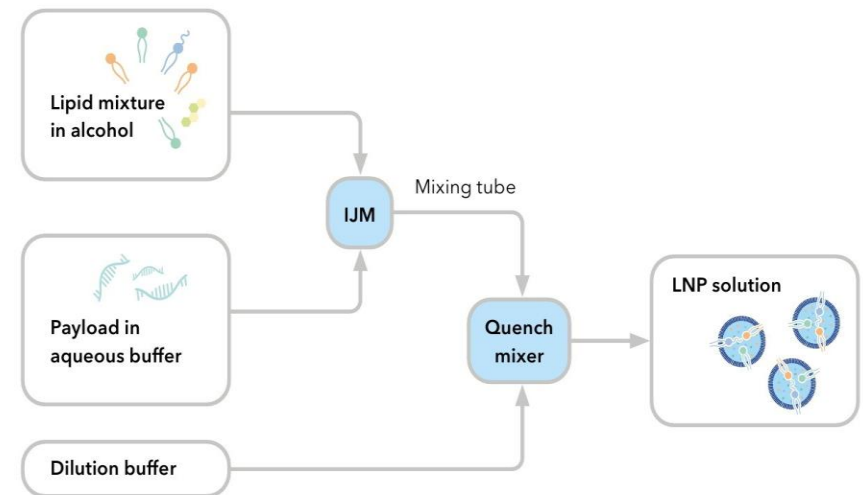
	Top-Down	Bottom-Up				
	High energy methods Extrusion / HPH / Sonication	Thin Film Hydration	Manual Batch / Pipetting	SCF Supercritical Fluids	Macrofluidics T-junction / IJM	Microfluidics Herringbone / Baffle
Size control & repeatability	Good	Low	Low	Good	Average	Excellent
Homogeneity	Average (multi steps)	Average	Low to Average	Low	Average to Good	Great (PDI <0.2)
Encapsulation efficiency	Average	Good	Average	Good	Good	Excellent (>95%)
Processing time	Variable	Slow	Fast	Slow	Fast	Very fast
Achievable volume range	mL / L	mL	μL / mL	mL	mL / L	μL / mL
Commercially available	Yes	Yes	Yes	No	Yes	Yes
Main characteristics	✓ Easily scalable No low volumes Payload alteration	✓ Widely available Poor NP control No large scale	✓ Simple & affordable Low EE% Low NP control	High temperature and pressure Not commercially available	✓ Large scale No low volume Payload alteration	✓ Best NP control ✓ Low volume Learning curve

Impingement Jet Mixing (IJM)

Impingement Jet Mixing (IJM) is a rapid, continuous mixing technology used for the industrial production of lipid nanoparticles (LNPs).

- Two liquid streams collide head on at high velocity
- Collision creates intense turbulence
- Mixing occurs in milliseconds under turbulent flow
- Lipid self-assembly is triggered by rapid solvent dilution

Impingement Jets Mixing Technology



What happens inside the IJM chamber

- The two streams impinge directly at the mixing point
- High Reynolds number flow generates strong turbulence
- Turbulence eliminates concentration gradients instantly
- Ethanol concentration drops rapidly → lipid solubility collapses
- Nucleation :- Small lipid clusters form uniformly throughout the volume
- Growth arrest :- Fast quenching prevents uncontrolled aggregation

Encapsulation mechanism

- Ionizable lipids become positively charged.
- Negatively charged mRNA binds electrostatically.
- mRNA is trapped inside the forming lipid structure.

Why does IJM gives narrow size distribution?

- Uniform nucleation dominates over growth.
- No local concentration gradients.
- Particle size is fixed within the first milliseconds.

Result

- Lipids become supersaturated.
- Lipid molecules self-assemble.
- mRNA is electrostatically encapsulated during particle formation.
- Now LNPs are formed inside IJM and now we need to analyze (using PAT) it and filter it.

Key Concept :-

Mixing time is much shorter than nanoparticle assembly time, leading to uniform and reproducible LNPs.

Process Analytical Technology (PAT)

- Framework for real-time monitoring, analysis, and control of pharmaceutical manufacturing processes.
- It focuses on identifying and controlling:
Critical Quality Attributes (CQAs)

Critical Process Parameters (CPPs)

Why PAT is needed?

- LNP production is rapid and continuous
- Small variations in flow, mixing, or concentration can affect:
 - Particle size
 - Encapsulation efficiency
 - Stability
- PAT ensures consistent quality during production, not just end-product testing

Key CQAs monitored

- Particle size (< 200 nm).
- Polydispersity index (PDI < 0.3).
- Zeta potential ($\approx \pm 30$ mV).
- Encapsulation efficiency ($> 80\%$).

Key CPPs monitored

- Flow rate and flow ratio.
- Mixing intensity/pressure.
- Temperature.
- Solvent (ethanol) concentration.

□ PAT continuously monitors critical quality attributes and process parameters to ensure consistent LNP quality during manufacturing.”

Quenching

Quenching by dilution is the immediate dilution of the freshly formed LNP suspension with an excess aqueous buffer.

- First , LNPs are formed.
- Now, quench buffer is added which results in sharp drop of ethanol concentration which reduces lipid solubility and mobility.
- Finally, particle growth ceases, preventing further fusion or aggregation, thereby locking in the final particle size and PDI.

Quenching by filtration is performed using Tangential Flow Filtration (TFF) or Membrane-based filtration to remove:

- Residual solvent (ethanol)
- Unencapsulated components

THIS IS THE FINAL STEP OF LNP MANUFACTURING.

NxGen Mixing

- NxGen mixing is a controlled rapid-mixing technology.
- Based on controlled nanoprecipitation of lipids with nucleic acids.
- Mixing is done under high controlled flow conditions.
- Precise control over mixing time and flow profiles enables uniform particle formation and low PDI.
- Scale-up achieved via parallelization of microfluidic cartridges
- Single-use fluid paths reduce cross-contamination risk

RESOURCES

- <https://insidetx.com/resources/reviews/lip-synthesis-overview-of-the-manufacturing-methods/>
- https://en.wikipedia.org/wiki/Lipid-based_nanoparticle
- https://drive.google.com/drive/folders/10AyL6bArZ2kxkyHl5krHhYheAy40hMO?usp=drive_link

THANK YOU!!